

PORTFOLIO · 2026

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Manufacture. Test. Analyze.

Hands-on semiconductor and thin-film process work, from cleanroom deposition and lithography to electrical testing and diagnosing why a device behaved the way it did.

Astronautical Engineering, Junior

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ExSTENT 2026 · UC San Diego cleanroom program



One process, end to end

Across four projects I worked every stage of the loop: make the device, measure it, and work out why it behaved the way it did.

01 MANUFACTURE

- Spin-coated and annealed a five-layer perovskite solar cell inside a glovebox
- Sputtered niobium thin films and patterned wafers with contact lithography
- Soldered and assembled a distortion-pedal circuit
- Modeled a two-part diagnostic cassette in CAD

02 TEST

- Forward and reverse current-voltage sweeps with a source-measure unit
- Resistivity of patterned niobium wires, with reversed current polarity
- Stylus profilometry and optical microscopy of films
- Fiber-coupled spectrum measurement of a photonic chip

03 ANALYZE

- Extracted Voc, fill factor and power output from J-V data
- Caught a non-physical efficiency result and flagged it instead of reporting it
- Compared resistivity across wire widths and linked results to process variables
- Critiqued my own pedal build for its weak solder points

THE FOUR PROJECTS



01 Perovskite solar cells

p. 3-4



02 Niobium thin films

p. 5-7



03 Integrated photonics

p. 8



04 Biotech and semiconductors

p. 9



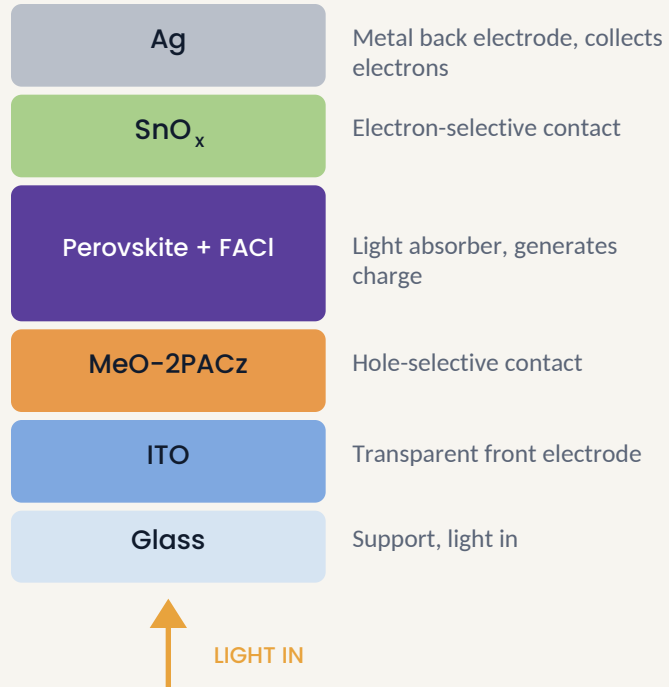
05 Distortion pedal build

p. 10

Building the cell

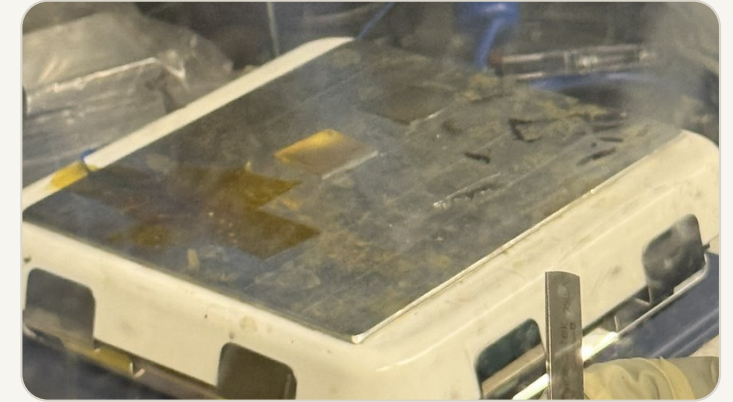
Five functional layers on glass, built with spin coating, annealing and evaporation.

DEVICE STACK



PROCESS FLOW

- 1 UV-ozone, 20 min**
Cleans ITO, improves coating
- 2 MeO-2PACz, 200 µL**
Static spin coat, 100 °C for 10 min
- 3 Perovskite (FAPbI₃ + FAPbI₃ + FAPbI₃), 200 µL**
Static spin coat, 100 °C for 10 min
- 4 SnO_x, 200 µL**
Dynamic spin coat, 150 °C anneal, 10 min
- 5 Ag electrode**
Thermal evaporation
- 6 Electrical testing**
J-V sweeps, forward and reverse



Substrates on the hotplate for annealing.



Working in the glovebox. The perovskite is air- and moisture-sensitive.

WHAT I HAD TO CONTROL

Surface cleanliness

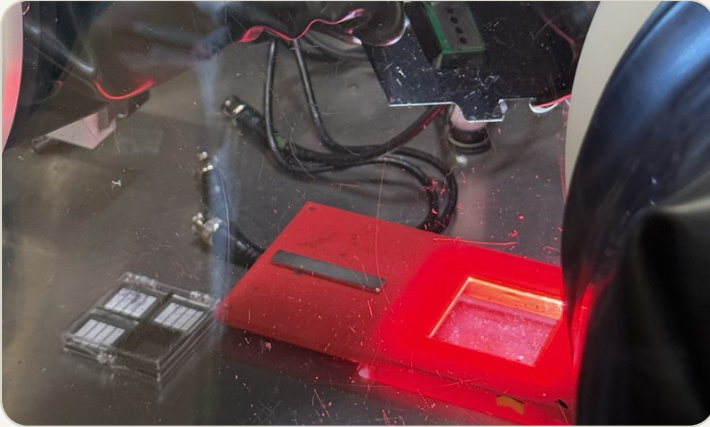
Full, smooth coverage

Spin speed and volume

Air and moisture

Testing, and finding out why

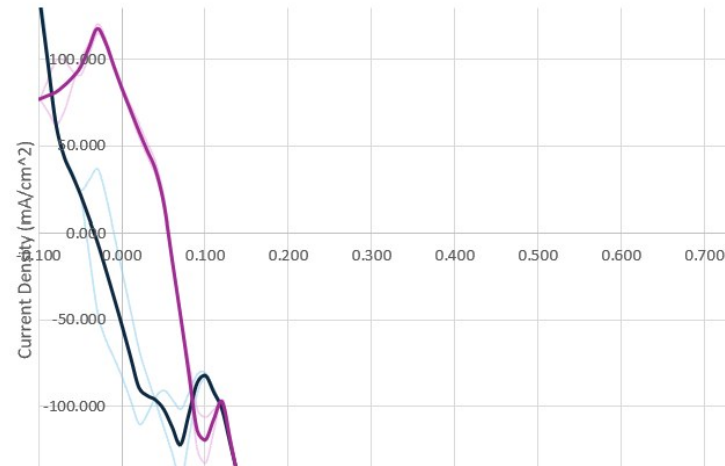
Best device: Cell 3, Pixel 1 (C3_P1). Reverse sweeps gave a real photovoltaic response; forward sweeps never formed a power-producing quadrant.



Device under test in the measurement fixture.



Bench source-measure unit used for the sweeps.



Averaged J-V sweeps for C3_P1, forward and reverse.

SYMPTOM / LIKELY CAUSE (MY HYPOTHESES)

Low Voc Recombination at defects or incomplete film coverage

Low fill factor Series resistance from uneven layers

~1.4% efficiency Fabrication variability plus electrical losses

C3_P1, REVERSE SWEEP

0.056 V Open-circuit voltage

0.295 Fill factor

1.39 mW/cm² Max power density

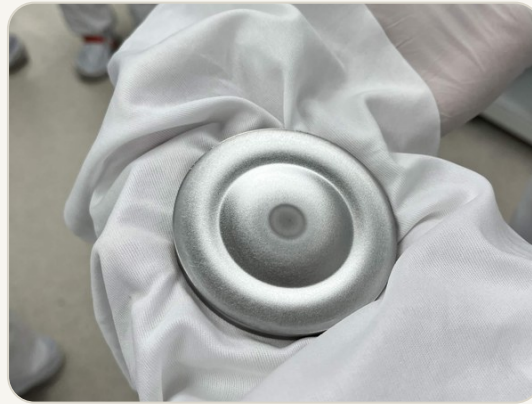
≈ 1.4% Power conversion efficiency

DATA-QUALITY CATCH

The raw extraction returned J_{sc} of 84 mA/cm² and PCE above 100%. Both are non-physical, caused by the very low Voc and a single integration interval. I flagged it instead of reporting it.

Depositing niobium films

DC sputtering of niobium on a Denton Discovery system, run from the cleanroom control cabinet.



Niobium sputtering target.



HOW A RUN GOES

- 1 Prepare**
Gowned cleanroom handling of the substrate, wiper and carrier to keep particles off the surface.
- 2 Load**
Mount the wafer on the carrier and place it on the stage under the sources.
- 3 Select the source**
Choose the niobium target on the cathode panel and set DC sputter mode on the touchscreen, checking the interlocks.
- 4 Deposit**
Sputter the film onto the wafer; the target erodes in a ring that shows where material comes from.
- 5 Unload and inspect**
Check the mirror-like film before patterning and measurement.

DC sputtering

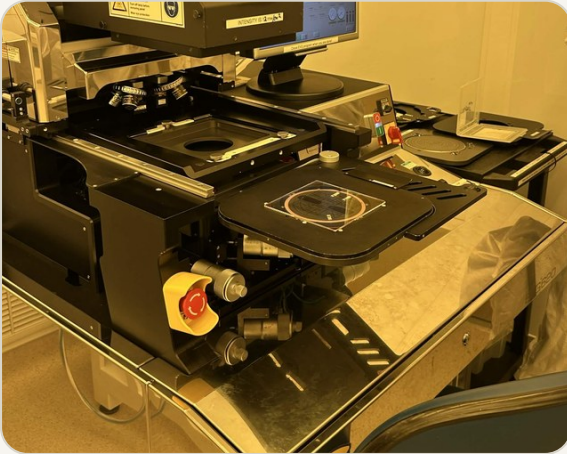
Denton Discovery

Nb target

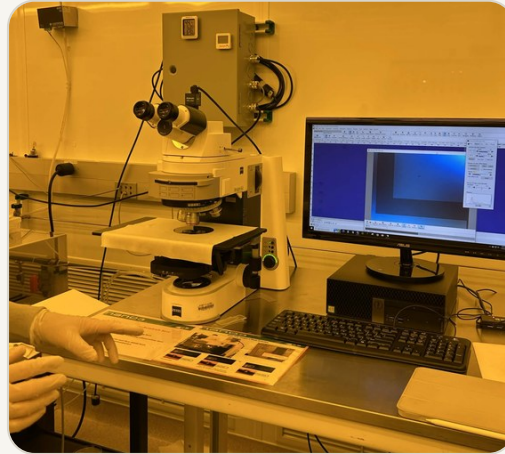
Cleanroom protocol

Patterning and measuring the film

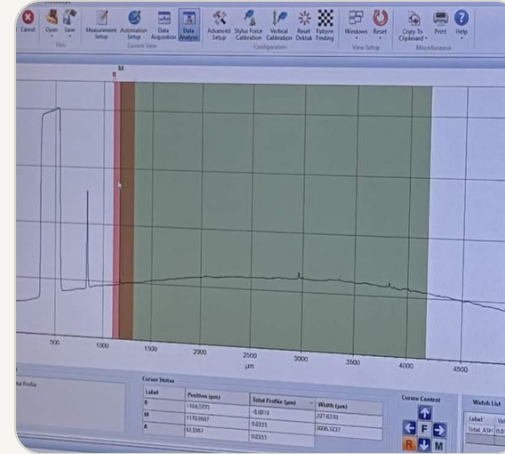
Contact lithography, microscopy and stylus profilometry to define, check and compare films.



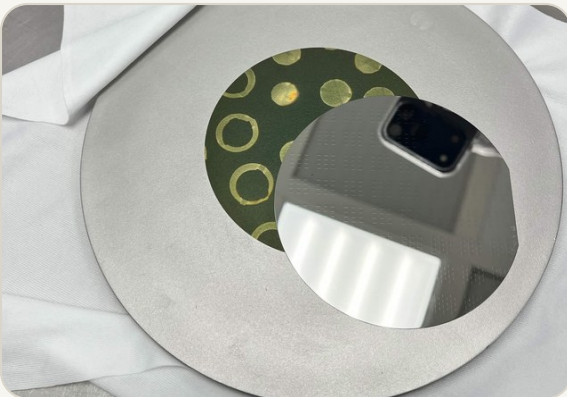
EVG 620 mask aligner in the yellow-light room.



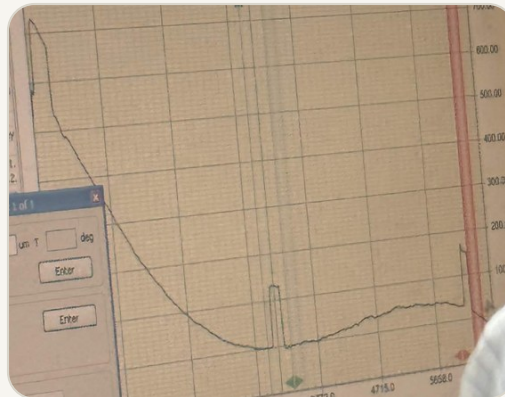
Microscope station for inspecting patterns.



Bruker profilometer scan across a film edge.



Patterned wafer on the substrate carrier.



Profile trace used to compare patterns.

WHY THIS STEP MATTERS

Thickness and pattern quality set a film's resistance, so profilometry closes the loop between deposition settings and electrical behavior.

Pattern

EVG 620 mask aligner, UV source rated at 12 mW/cm^2 , features defined through a mask.

Inspect

Zeiss microscope with camera software to check pattern fidelity.

Measure

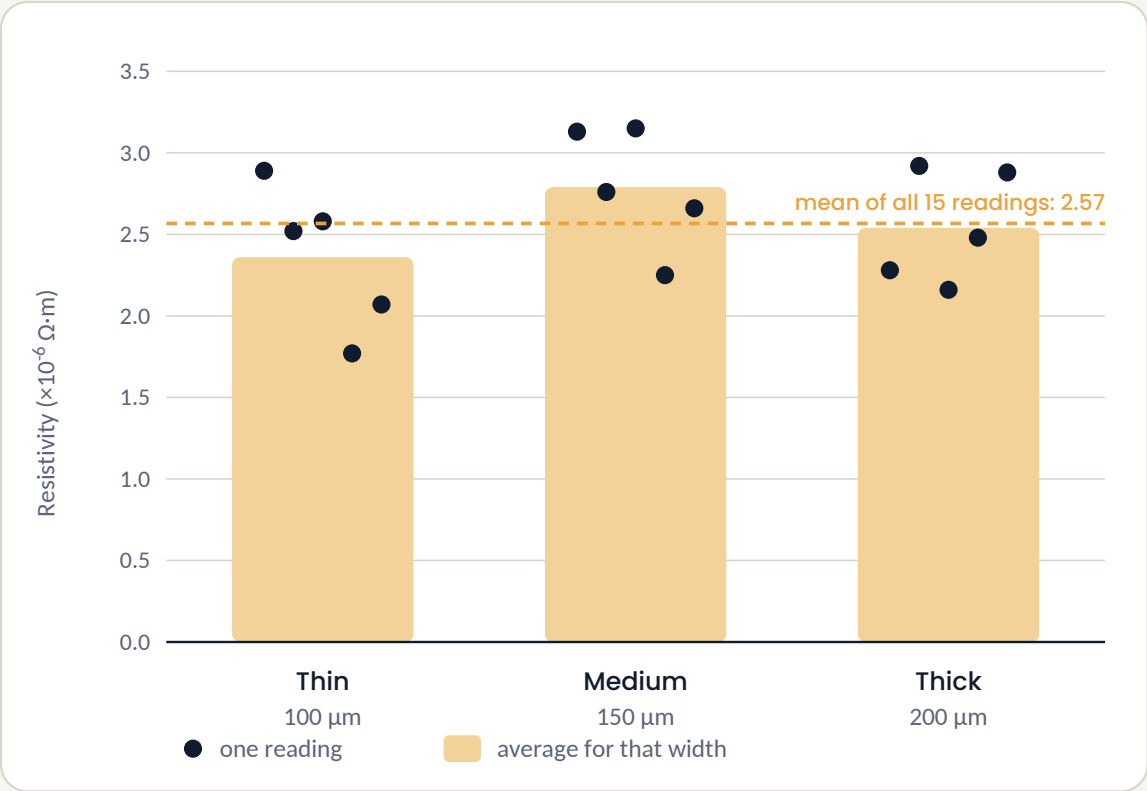
Stylus profilometry: 6.6 mm scans at 5 mg stylus force, with step-height readouts.

Analyze

Compared resistivity across 100, 150 and 200 μm wires (next page).

Resistivity across wire widths

One film, three patterned wire widths. If the film is uniform, resistivity should come out the same for all of them.



Wire	Avg resistance	Avg resistivity (10 ⁻⁶ Ω·m)
Thin, 100 μm	364 Ω	2.36
Medium, 150 μm	286 Ω	2.79
Thick, 200 μm	196 Ω	2.54

ρ = R · w · t / L, with film thickness t = 130 nm and probe spacing L = 2 mm

WHAT THE DATA SAYS

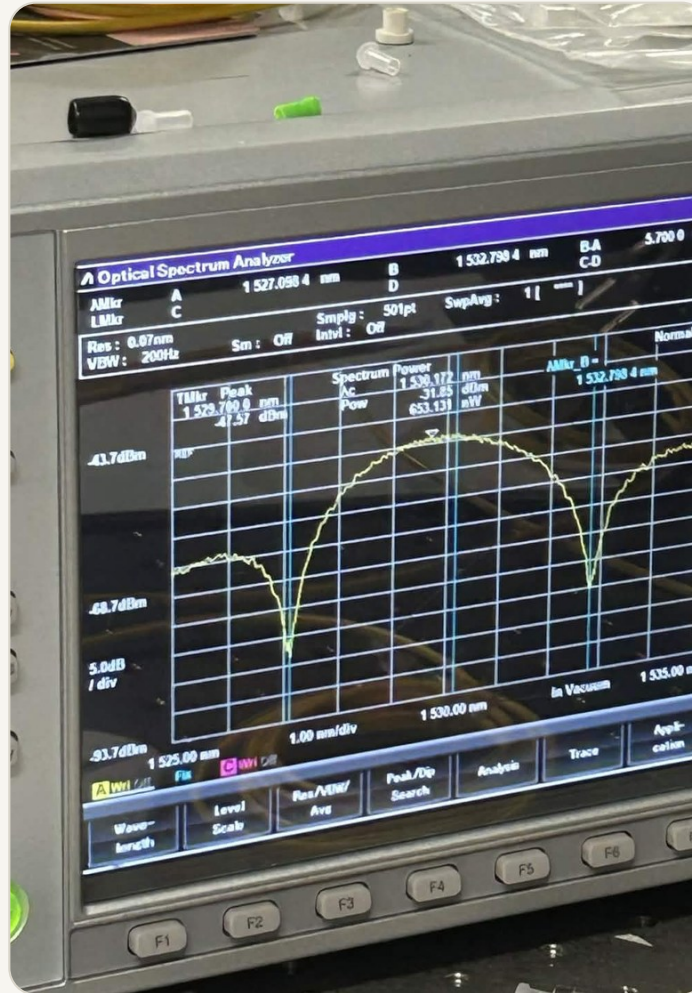
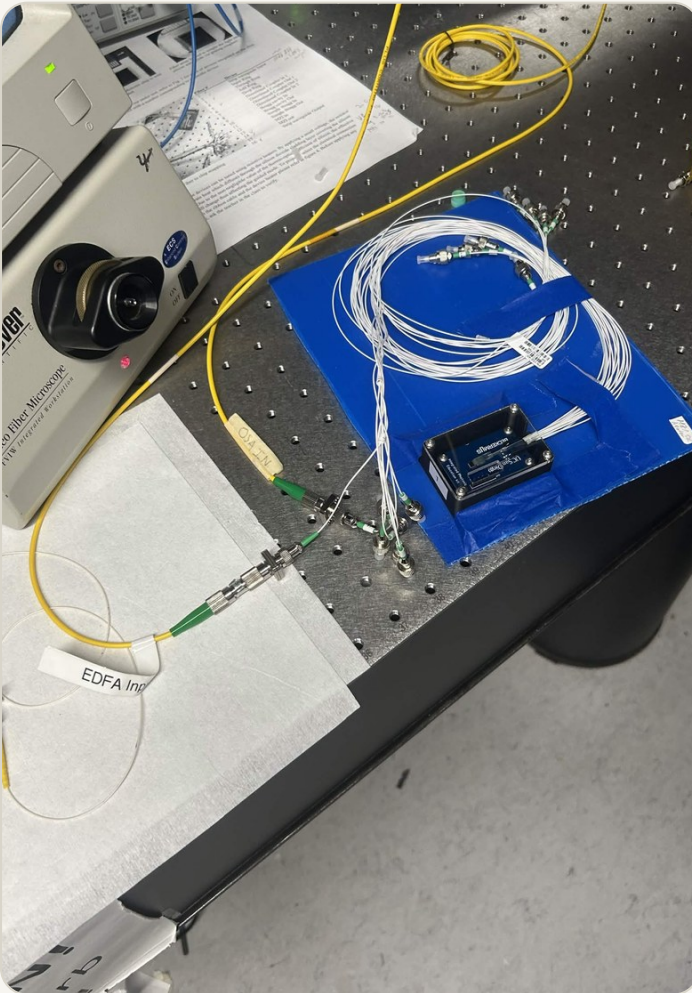
- Resistivity lands at 2.4 to 2.8 ×10⁻⁶ Ω·m across the three widths, within about ±10% of each other. Resistance falls as the wire widens (364 to 196 Ω) while resistivity holds roughly steady, as it should for one film.
- Scatter within a single width is larger (thin wires ranged 1.8 to 2.9), which points to film or contact non-uniformity.
- Open question: these values are about 17 times bulk niobium (roughly 1.5 ×10⁻⁷ Ω·m). Oxygen content, grain structure and thickness accuracy are the first things to check.

MEASUREMENT SETUP

- Film 130 nm thick
- Voltage probes 2 mm apart
- ±2 mA, 5 readings per width

Characterizing a photonic chip

Light in through fiber, out to a spectrum analyzer, and a resonant response you can read directly off the trace.



SETUP

Fiber-coupled integrated photonics chip on an education kit, fed through an EDFA input and read out on an Anritsu MS9740B optical spectrum analyzer, with a fiber microscope station for inspecting connectors.

WHAT THE TRACE SHOWS

1525–1535 nm

Scan window

0.07 nm

Resolution

5.7 nm

Spacing between the two dips

Periodic dips in transmission are the signature of a resonant structure. Their spacing gives the free spectral range, a key design parameter for filters and modulators.

Where biotech meets chip manufacturing

ExSTENT 2026 team poster, plus the cassette I modeled for it.

Biotechnology
 Sarah Leano, Jamie Miller, Min Nguyen, Yaretsi Rubio, Edward Torres

Why Biotechnology Matters

- Biotechnology combines biology and engineering to create diagnostic tools and healthcare technologies.
- Semiconductor sensors and lab-on-a-chip devices integrate sensing, fluid handling, and analysis for smaller, faster diagnostic testing.

Fig 1. An example of a lab-on-a-chip diagnostic device.

Scientific Foundations: Biomolecular Detection

- DNA probes bind to complementary DNA sequences, allowing specific genetic targets to be identified and analyzed.
- Antibodies bind to specific antigens, helping isolate biological targets and convert their presence into a detectable signal.
- Specific molecular interactions allow biological targets to be selectively recognized and detected.

Fig 3. Antibodies bind to specific antigens, allowing biological targets to be recognized and detected.

Fig 4. DNA probe hybridization enables specific genetic detection and analysis.

DNA Analysis: Gel Electrophoresis

How it works?
Gel electrophoresis separates PCR-amplified DNA from one STR region by separating using gel electrophoresis. Band positions were compared to a DNA ladder to estimate fragment size.

Fig 2. Set up of Gel electrophoresis by BOSTONind.com

DNA is negatively charged due to its phosphate backbone. When an electric field is applied, DNA migrates through the agarose gel toward the positive electrode (+).

What DNA separation demonstrate?

STR variation: Different people have different numbers of repeats.

Different fragment lengths: More repeats = larger fragments; fewer repeats = shorter fragments.

Gel separation: Fragments separate by size during electrophoresis.

Band pattern: Each sample shows a unique pattern of bands.

DNA comparison: Band patterns are compared to determine if samples match or do not match.

Lateral-Flow Assays: How They Work

LFA convert a specific antibody-target interaction in a rapid and visible result.

Lateral-flow assay: Capillary action moves the sample through the strip. Target molecules bind labeled antibodies and are captured at the test line, producing a visible sign. The control lines confirms proper flow. Adapted from Sachdeva et al.(2021)

Engineering a Diagnostic Device

From Lab to Industry: CorDx

Lab Skills	Manufacturing	Semiconductors
Measure & Prepare	Control & Scale	Sense & Process
Pipetting → Precision	Process Control → Consistency	Biosensors → Detection
Centrifuge → Separation	Quality Control → Reliability	ICs → Signal Processing
pH → Controlled conditions	Scale-Up → Mass Production	Automation → Digital Results

Connecting Biology, Engineering, Manufacturing

Biology → Engineering → Manufacturing → Diagnostics

Biology identifies the target, engineering creates the detection system, and manufacturing produces reliable devices at scale. Together, these fields turn biological discoveries into practical diagnostic tools.

References & Acknowledgments

- Sachdeva et al. (2021). Frontiers in Bioengineering and Biotechnology, 8, 602659. <https://doi.org/10.3389/fbioe.2020.602659>

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The program is made possible through generous collaborations with CorDx, First Solar, Quantum Design, and the Quantum Optical Institute of Southern California.

The idea

Biotech diagnostics rely on semiconductor sensors and manufacturing discipline. The poster follows biology to engineering to manufacturing to diagnostics, framed by CorDx.

Team experiment

PCR-amplified DNA from one STR region, separated by gel electrophoresis and compared to a DNA ladder to estimate fragment size.

My part: the cassette

Two-part CAD model: a base tray with a recessed strip bed, and a lid with a read window, sample port, C/T/S markings and a flow arrow.

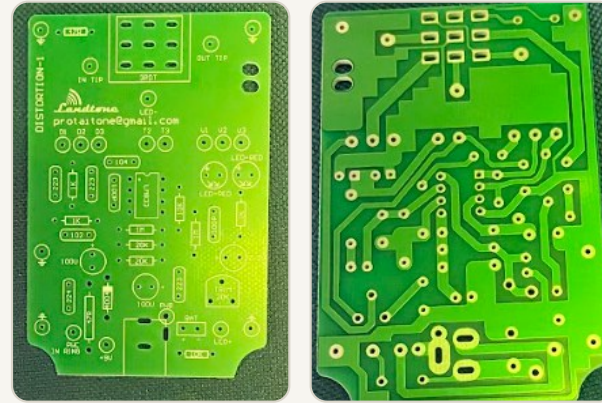
THE PROGRAM

ExSTENT 2026: Experiential Semiconductor Training in Emerging NanoTechnology, at UC San Diego. NSF ExLent Beginnings funded, with CorDx, First Solar, Quantum Design and the Quantum Optical Institute of Southern California.

Team poster (five authors). The Engineering a Diagnostic Device panel shows my CAD cassette.

Building a distortion pedal

E&M; honors physics project: soldering the circuit, and understanding why clipping makes distortion.



Bare PCB, front and back.

How it makes distortion

Turn the gain up until the signal exceeds what the circuit can pass and the wave tops get clipped flat. Diodes do the clipping once the signal passes their forward voltage, and the reshaped wave adds new frequencies we hear as distortion. Silicon diodes clip near 0.7 V, LEDs at 1.8 to 3.3 V, which gives a tighter, more dynamic crunch.

Diodes

The clippers: silicon or LED.

Resistors

Set gain and distortion amount.

Capacitors

Filter noise and shape the tone.

Potentiometers

Gain, tone and volume knobs.

MY DESIGN CRITIQUE

The footswitch attaches directly to the PCB, so a sudden knock can stress and crack solder joints. Wiring the switch to the board adds joints, but takes shock off the ones that matter, lowering overall risk of failure.

What I can do, and on what

Fabrication

- Static and dynamic spin coating
- Hotplate annealing
- UV-ozone surface treatment
- Thermal evaporation of Ag
- DC sputtering of Nb
- Contact lithography
- Through-hole soldering and circuit assembly

Characterization

- J-V sweeps with a source-measure unit
- Stylus profilometry, step height
- Optical microscopy
- Optical spectrum analysis
- Fiber-coupled device testing
- Resistivity of patterned wires

Analysis

- Voc, fill factor, Pmax, PCE extraction
- Forward vs. reverse scan comparison
- Root-cause reasoning from data
- Sanity-checking non-physical output
- Resistivity across wire widths
- Design critique of failure points
- Research posters and write-ups

Environments and tools

- Cleanroom gowning and contamination control
- Glovebox handling
- Denton Discovery sputter system
- EVG 620 mask aligner
- Bruker Vision64 profilometer
- Anritsu MS9740B OSA
- CAD modeling
- Gel electrophoresis basics

WHAT THE WORK TAUGHT ME

Reproducibility was the hard part. Fabrication quality drove device performance, and process control matters as much as materials.